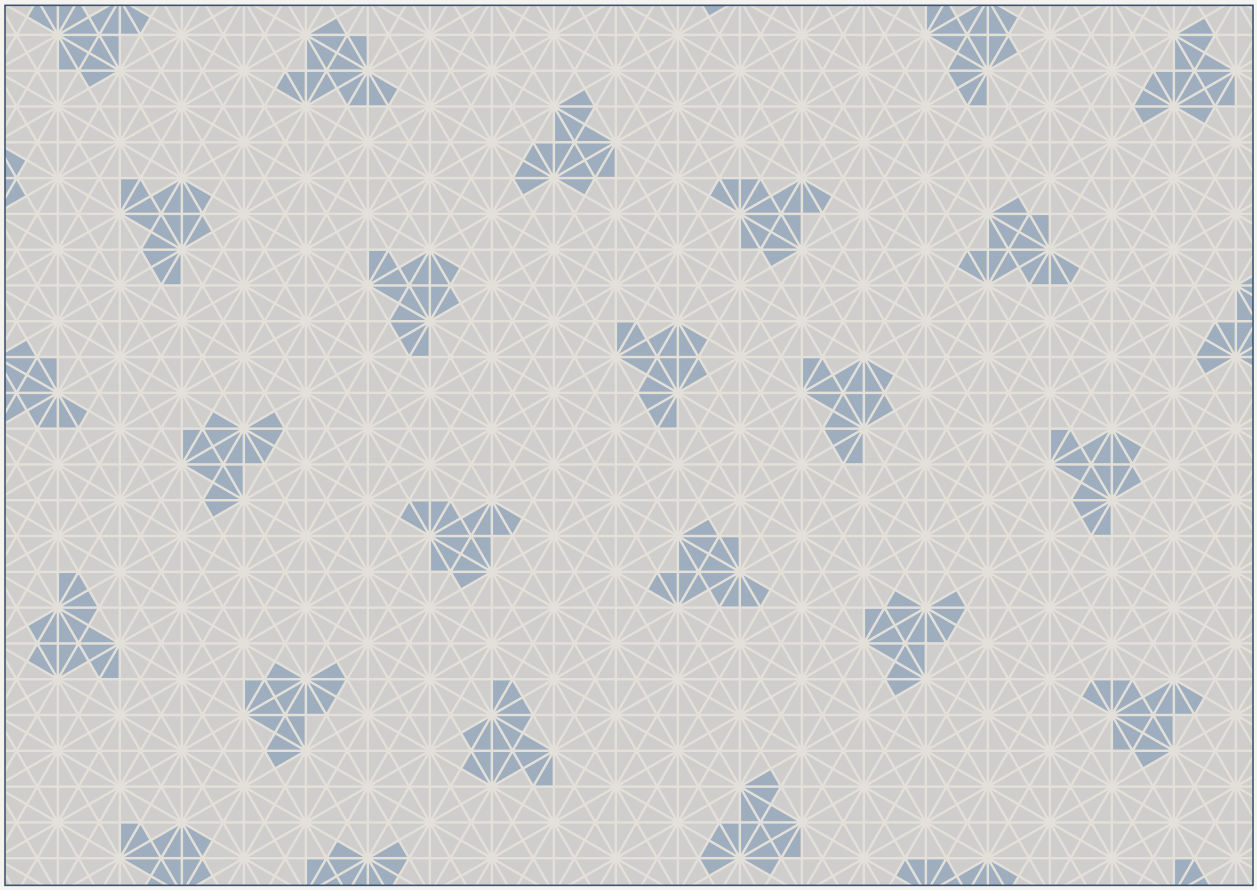
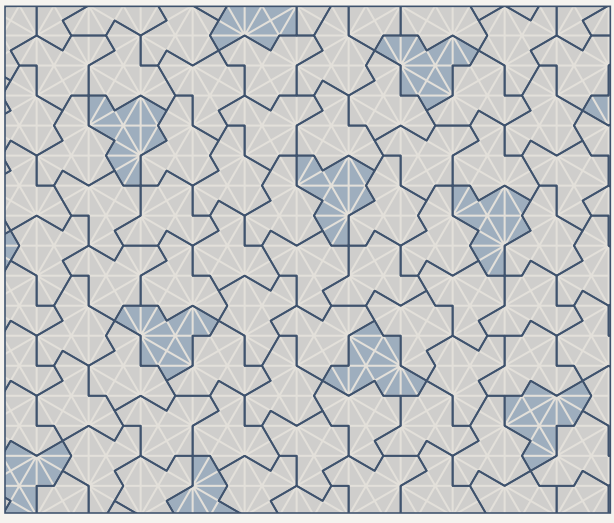
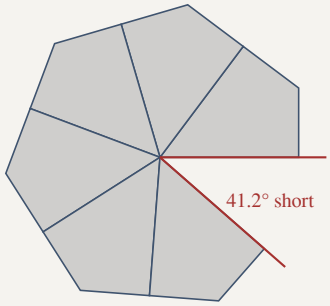
An “einstein” wall: hats from diagonally cut tiles



as laid: reflected hats blue-grey, all others light; uniform grout between every piece



the same, with the hats outlined

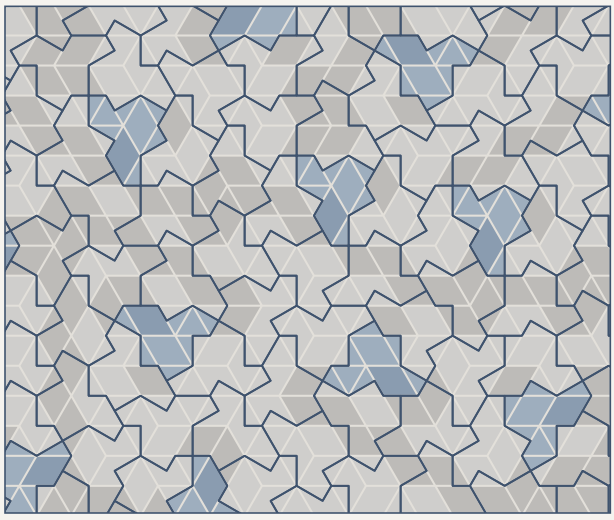


Six kites made from 1 : 2 triangles meet at only 318.8° (6 × 53.13°), not 360°. A hat kite needs triangles with legs 1 : √3, angles 30° and 60°.

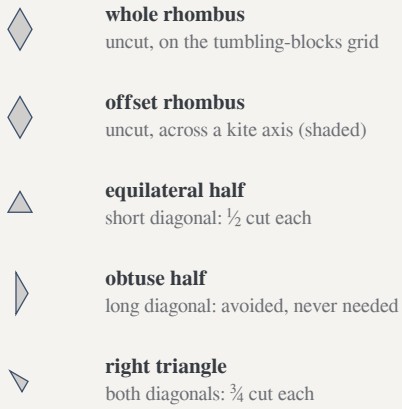
The same wall in 60° rhombus tile



laid for the fewest cuts: whole rhombi almost everywhere, cut pieces only around the blue hats



hats outlined; rhombi laid across a kite axis shaded



whole rhombus
uncut, on the tumbling-blocks grid

offset rhombus
uncut, across a kite axis (shaded)

equilateral half
short diagonal: ½ cut each

obtuse half
long diagonal: avoided, never needed

right triangle
both diagonals: ¾ cut each

Within 30 units of the centre (804 hats): 992 cuts, against 1,374 using only the tumbling-blocks grid and 6,432 cutting every half-kite from 1 : √3 rectangles. Each blue hat is two whole rhombi, one equilateral half and six right triangles: 5 cuts. The light field takes about six more right triangles per blue hat. The shaded chains of offset rhombi link the blue hats in pairs, so the light tile closes up around each hat without further cuts. Cuts count ½ per half and ¾ per right triangle.